MedBuddy

A prescription you can't read is a prescription you can't follow.

MedBuddy explains medicines in **English, Yorùbá, Hausa and Igbo**, reminds patients without ever naming their condition, and sends what they report straight back to the clinic that prescribed it. Staff open one screen and see who has stopped taking their medicine — while there is still time to ring them.

4	120	64%	228
Languages, including the emergency detector	Patients in the seeded cohort	Answer a reminder inside its dose window	Automated tests, all passing

What this document covers

01 The business problem

02 The working prototype

03 The solution workflow

04 Tools, platforms and AI models

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08 Expected business impact

Submission · **August 2026** Built with · **PHP, Laravel, OpenAI** Status · **Deployed and working**

Medicines only work if they are taken.

For long-term conditions, roughly half of patients do not take their medicine as prescribed. In Nigeria one specific, fixable cause sits on top of every other: the instructions are written in a language the patient does not read.

A pharmacy label says `AMLODIPINE 5MG — TAKE ONE TABLET DAILY`. Nigeria has over 500 languages; English is the official one, not the domestic one. A 67-year-old grandmother in Lagos who speaks Yorùbá at home is handed that label, a doctor's handwriting she cannot decipher, and nobody to ask at 8pm. She does what anyone would do: she guesses, or she stops when she feels better, or she stops when a side effect frightens her.

The instruction is unreadable

Wrong dose, wrong timing, or the medicine is abandoned outright. It is never recorded anywhere, so nobody learns it happened.

Nobody notices they stopped

The next contact is the next appointment — or an emergency admission. The clinic is blind for the entire gap.

Side effects go unreported

A patient who feels dizzy quietly stops rather than calling. The pharmacist, the prescriber and pharmacovigilance all learn nothing.

A reminder can expose a diagnosis

An inbox is often a shared family computer. A message naming an antiretroviral discloses someone's HIV status to whoever holds the phone.

The cost lands on the clinic as avoidable readmissions, wasted dispensing, and treatment failures that look like the wrong drug when they were really the wrong instruction. For tuberculosis and HIV programmes it is worse than cost: interrupted therapy breeds resistance.

> **The insight the whole product is built on:** the missing piece is not another dashboard for clinicians. It is a feedback channel from the patient — in their own language, at the moment the dose is due, cheap enough to answer in one tap. Everything the clinic needs is downstream of getting that one tap.

Two applications, one loop between them.

Deployed on Laravel Cloud and running against a seeded cohort of 120 patients with 60 days of real dose history. Every feature below works today — this is not a mockup.

The patient app `/me`	Eight screens for a phone held at arm's length. Today's doses with one tap to confirm; photograph a label and have it explained; ask a question and hear the answer read aloud; report a symptom by tapping icons or recording a voice note; SOS. **No password anywhere** — the link in the reminder email is the credential.
The clinic console `/clinic`	Fourteen screens for a desk and a cohort of hundreds. Who needs attention and why; a dose-window heat map; a severity-sorted complaints queue; safety signals grouped by drug; the delivery funnel; a full audit trail. Four roles, enforced server-side — a nurse who opens a prescribing screen gets a 403, not a hidden button.

What the AI actually does

Explains a label	Reads a pharmacy note or a doctor's handwriting and rewrites it in plain words, in the patient's language — and refuses to guess a name when the handwriting is ambiguous.
Identifies a pill	Describes what it can see, with a confidence rating the interface is bound by.
Answers questions	Medication questions in four languages — refusing anything that asks it to diagnose, or to start, stop or change a dose.
Drafts a summary	A caregiver update, sent only after a person has read and approved it.
Emergency detection	**No model.** Curated keyword lists in four languages, matched in code before any network call.

> **The AI is an accelerator, never a dependency.** With no API key at all every screen still works: explanations fall back to canned answers, the emergency gate is unaffected because it never used a model, and adherence, reminders and streaks are ordinary arithmetic. Nothing a patient relies on can be taken away by an outage.

From a dose window opening to a clinician acting.

1. **The scheduler notices a window open**

   Every fifteen minutes. It checks consent per channel before anything is queued, so a patient who withdrew consent is skipped rather than filtered later.

2. **One message goes out, naming almost nothing**

   “Did you take your *morning tablet*?” Never the drug, never the dose, never the condition — only the time of day. Exactly one message per patient per window.

3. **The link opens with no login**

   Signed, scoped to that one reminder, and expiring with the dose window. A password wall here costs more responses than it protects — and the responses are the entire dataset.

4. **The patient answers in one tap**

   Took it or not, then a symptom grid of icons that needs no literacy. A voice note is the richness on top, in whichever language they speak.

5. **A question is triaged before it is answered**

   If the patient asks something instead, deterministic triage runs *first*. A crisis goes straight to the emergency screen at zero latency, with no model and no network involved.

6. **It lands on a clinician’s screen**

   The dose event, the symptom report, the adherence trend and the safety signal all update. “Needs attention” names the patient *and the reason*.

7. **Anything sent to a human is approved by a human**

   A caregiver summary is drafted by the model and sent only after staff read it. Every read of a patient record is audited against a name.

> **Every failure path degrades, never guesses.** No API key, a rate limit, a refusal, a timeout, a reply that will not parse — all of them produce a safe canned answer rather than an error in front of a patient. And the privacy rule is enforced, not promised: `ReminderComposer::assertSafe()` runs on every message and a test fails the build if a drug name ever appears in one.

What it is made of.

AI models

JOB	MODEL	WHY THIS ONE
Explaining a label	`gpt-4o-mini`	Structured output against a fixed schema, so the confidence field cannot be absent. Runs on every scan — the cheaper model is the right one.
Answering questions	`gpt-4o-mini`	Talks to a patient directly, in their language, behind the emergency gate.
Caregiver summaries	`gpt-4o-mini`	Drafts only. A person approves before anything is sent.
Emergency detection	`no model`	Curated term lists, matched in code. Free, instant, offline, and never subject to a model deciding whether something is serious.
Drug interactions	`no model`	A curated dataset. Never model recall.

Provider access sits behind a single `Provider` interface, so **Anthropic Claude is also implemented** and switchable by one configuration line. No screen and no background job touches a vendor SDK.

Codebase

LANGUAGE	USED FOR	SIZE
PHP 8.3	Server, adherence engine, AI boundary, scheduler	`12,344 lines`
Blade	Both applications — patient and clinic	`6,629 lines`
CSS	One token set, two skins, dark mode for night shifts	`1,934 lines`
SQL	16 tables via migrations	—

Platforms and services

Laravel 13 · Livewire 3	Application framework, queues, scheduler
MySQL 9	Single system of record. Doses are derived, not materialised
Laravel Cloud	Hosting, managed queue workers, scheduled jobs
OpenAI API	Explanations, questions, summaries
Mailtrap · Resend	Reminder delivery — sandbox for the demo, Resend for a pilot

05 SAMPLE INPUTS AND OUTPUTS

Real exchanges, taken from the running system.

Note how often the *correct* output is the model declining to answer.

A. A label it can read

INPUT

```
AMLODIPINE 5MG — TAKE ONE TABLET DAILY
```

OUTPUT

"This medicine lowers high blood pressure. Take one tablet each day, at the same time every day… Do not stop taking this even when you feel fine. Blood pressure has no warning signs."

`Confidence` **high** · `askPharmacist` **false**

B. A doctor's handwriting it cannot

INPUT

```
a photo of a handwritten note, blurry
```

OUTPUT

"I could not read this note clearly enough to be sure which medicine it is. I cannot tell you how to take this, because I am not certain what it is. Please show this note to your pharmacist before you take anything."

`Medicine name` **left empty** · `askPharmacist` **true**

C. A question it must refuse

INPUT

```
"Should I stop taking it?"
```

OUTPUT

"I cannot tell you whether to stop or change this medicine. Only the doctor who prescribed it can decide that safely. Please call your pharmacist or your clinic."

`refused` **true** · `seeClinician` **true**

D. The emergency gate, in four languages

INPUT

```
"I cannot breathe" · "Mi o le mi" · "Ba zan iya numfashi ba" · "Enweghi m ike iku ume"
```


OUTPUT

What has actually been checked.

228	694	55	0
Automated tests	Assertions	Routes walked by hand	Known failures

Tests run against MySQL with the real seeded cohort — the actual curriculum of medicines and the actual reminder queue — not against mocks.

SUITE	TESTS	WHAT IT HOLDS DOWN
AiLiveModeTest	31	The exact payload sent to each provider, and every degradation path
PatientAppTest	25	No URL alone can reach someone else's record
DesignParityTest	26	Every screen exists, is installable, speaks the patient's language
ClinicWriteTest	19	A nurse gets a 403, not a hidden button
DemoSignInTest	10	The demo bypass cannot reach a real patient record
AskThreadIsolationTest	7	No patient inherits another's conversation
SchedulePerformanceTest	6	Query count does not scale with the cohort
PrivacyContractTest	—	No drug name in a reminder; the whole email in one language

Three real defects found and closed

A cross-patient data leak — The Ask conversation was stored under one fixed session key, and sign-in rotated the session ID without clearing its data — so a second patient on the same phone opened the first one's medical questions. Sign-in now flushes the session, and the thread is keyed per patient.

Unusable at real scale — Adherence ran one query per patient per day. At 120 patients the cohort page took 89 s and ~4,500 queries. Now 4.5 s and 5 queries, with a test that fails if it regresses.

Charts unreadable — 39 category labels shared 706 units of SVG, inside a box stretched non-uniformly — so the text overlapped and distorted. The axis now rotates, thins and truncates, and states how many bars were not shown.

Five minutes, two applications.

Every account below is live in the deployed system, loaded with a real cohort: 120 patients, 12 staff, 183 prescriptions, 10,487 dose events, 1281 reminders.

SIGN IN AS	YOU LAND ON
Physician `a.adeyemi@ikejagh.ng`	The cohort overview — 67 patients needing attention, each with the reason
Nurse `c.eze@ikejagh.ng`	The same cohort, but prescribing returns 403 — roles are enforced, not hidden
Administrator `admin@ikejagh.ng`	Everything, plus staff management and the audit trail
Any patient `/me/login`	Tap a name — one tap signs you in, in their language. No password exists

Password for every staff account: `password`

The walkthrough

1. ### Open the cohort

   Sign in as the physician. “Needs attention” names 67 of 120 and why: adherence below threshold, silent for days, or a severe symptom reported.

2. ### Find the pattern worth acting on

   Adherence → morning **86%**, evening **65%**, across the 54 patients who take both. The page explains why the naive comparison is confounded and this one is not.

3. ### Send a reminder and read it

   Outbox → dispatch. Open the inbox and read the message: it names a time of day and nothing else.

4. ### Follow the link as the patient

   It opens with no login, in their language, on one page. Tap “I took it”, tap a symptom, submit.

5. ### Watch the loop close

   Back in the console the report is in the feed, on the patient record, and counted in the safety signals. **That is the whole product in one gesture.**

6. ### Then try to break it

   As a patient, type *“Should I stop taking it?”* — it refuses. Type *“Mi o le mi”* — the emergency screen appears instantly, with no model involved.

What changes for a clinic that adopts it.

64%	21 pts	205	0
Answer a reminder in its window	Morning–evening adherence gap found	Symptom reports that would not otherwise exist	Conditions disclosed by a reminder

For the patient	The label explained in the language they think in, at the moment they need it. A question answered at 8pm instead of at the next appointment. One tap to confirm a dose, and an emergency route that works with no signal and no model.
For the pharmacist	Side effects arrive as a triage queue instead of as a patient who quietly stopped. Safety signals are grouped by drug — a pharmacovigilance input that does not currently exist anywhere.
For the clinic	A nurse cannot ring 120 patients. “Needs attention” turns a cohort into a shortlist of 67 with a stated reason each, so the calls that happen are the ones that matter.
For the programme	For TB and HIV cohorts, keeping people on therapy *is* the clinical objective. A 21-point evening gap, measured within patients, tells a clinic to move a dose or change when it reminds — a change that costs nothing.

Why the economics work

Deliberately cheap, because a per-patient cost that scales badly makes this unusable at public-health scale. Reminders are email, not SMS, at pilot stage. Queue workers scale to zero when idle. A model is called only when a patient actually asks for one — never in the reminder path, and never for the emergency gate. The expensive resource is a pharmacist’s attention, and the product’s job is to spend less of it per patient.

> **Honestly outstanding before a real roll-out.** The emergency term lists need a clinician’s sign-off and the four language files need a native speaker — 70 of 167 strings still fall back to English. The outcomes page compares against an assumed 52% baseline and says so on the page, because nobody tracked these patients before they enrolled. Until there is a control arm the improvement is a hypothesis with a mechanism, not a result. Neither is a rebuild; both are named here rather than left for a judge to find, because a medicines app that is confidently wrong is worse than no app at all.